

Technical Documentation

GRZ Land Development Capacity Assessment SQL Framework

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Database: PostgreSQL with PostGIS

Primary schema: property

Main procedure: property.create_mv_property_grz_ldcap()

Primary purpose: Generate a repeatable spatial assessment of land development capacity within the **General Residential Zone (GRZ)**, including vacant land, infill opportunities, fully developed land, overlay constraints, and summary reporting.

1. Executive Summary

The SQL script implements a rule-driven land development capacity model for GRZ-zoned properties. It builds a chain of materialized views that assess properties against zoning, overlays, hazards, infrastructure constraints, building footprints, road frontage, access requirements, minimum lot-size rules, and spatial buildability tests.

The final outputs classify GRZ properties into:

- **Vacant viable**
- **Infill viable**
- **Fully developed**
- **District-wide summary**
- **Suburb-level summary**

The framework is designed to be regenerated through a single master procedure:

```
1 CALL property.create_mv_property_grz_ldcap();
```

This master procedure drops and recreates the dependent materialized views in the required processing order.

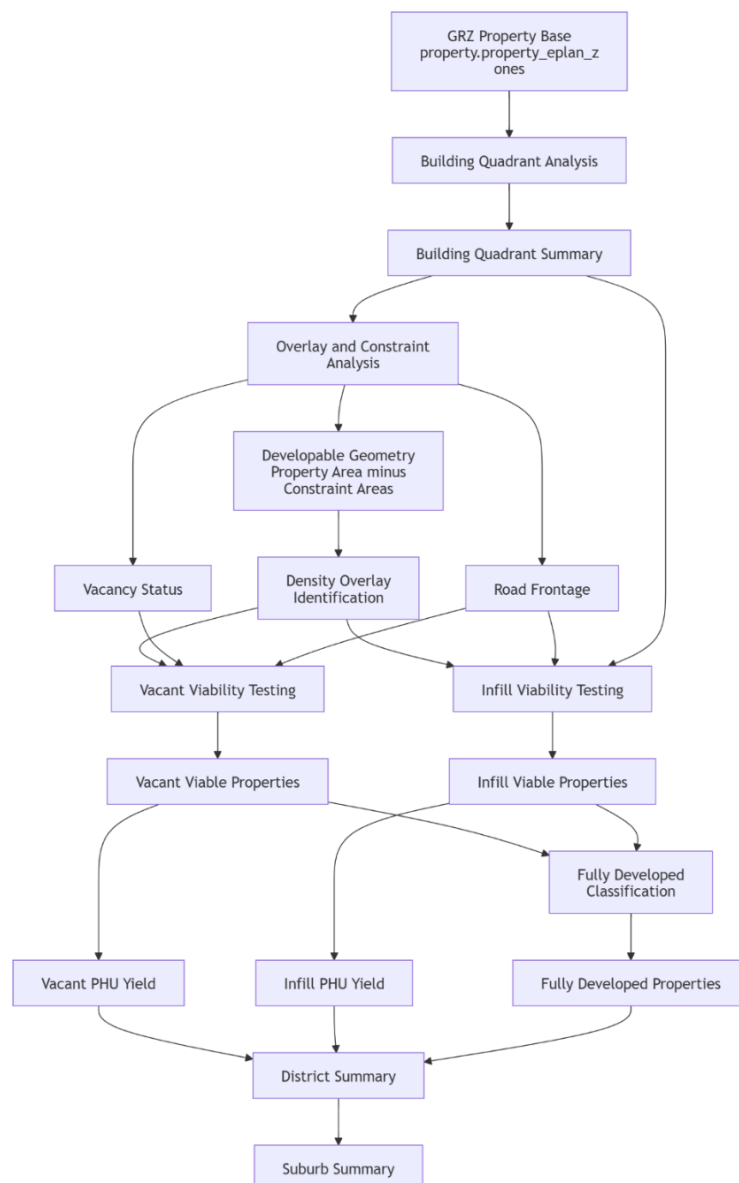
2. System Purpose

The GRZ LDCAP framework answers the following business questions:

1. Which GRZ properties have potential residential development capacity?
2. Which properties are vacant and suitable for development?
3. Which already-developed properties may support infill development?

4. Which properties are constrained by overlays, hazards, infrastructure, or inadequate access?
 5. What is the estimated potential household unit yield?
 6. How much capacity exists by development type, development scale, and suburb?
-

3. High-Level Process Flow



The SQL implements this flow through a sequence of stored procedures and materialized views.

4. Main Orchestration Procedure

Procedure

```
1 property.create_mv_property_grz_ldcap()
```

Description

This is the master rebuild procedure. It removes existing LDCAP materialized views and recreates them in dependency order.

Key Processing Order

```
1 1. property_grz_ldcap_building_quadrant_stats
2 2. property_grz_ldcap_building_quadrant_summary
3 3. property_grz_ldcap_viability_overlay_analysis
4 4. property_grz_ldcap_overlays
5 5. property_grz_ldcap_infill_viability_testing
6 6. property_grz_ldcap_vacant_viability_testing
7 7. property_grz_ldcap_fully_developed_testing
8 8. property_grz_ldcap_infill_viable
9 9. property_grz_ldcap_vacant_viable
10 10. property_grz_ldcap_fully_developed
11 11. property_grz_ldcap_summary
12 12. property_grz_ldcap_summary_suburbs
```

This order is important because later views depend on outputs created earlier in the process.

5. Core Input Datasets

The framework depends on several spatial datasets.

Dataset	Purpose
property.property_eplan_zones	Identifies GRZ-zoned property geometries.
property.property	Base property/cadastral geometry layer used for area, geometry, exclusions, and joins.
property.building_footprints_2025	Existing building footprints used for vacancy, infill, and coverage checks.
property.property_road_frontage	Road frontage and access assessment.

property.property_eplan_zones_multiples	Used to identify properties where GRZ is not the dominant zone.
general.whanganui_local_facilities	Used to exclude facilities such as health, schools, rest homes, prisons, marae, religious activities, and fuel facilities.
eplan.osz_open_space_zone	Used to exclude Open Space Zone properties.
property.property_building_consents_dwelling_gt20250206	Excludes properties with recent dwelling consents after the 2025 aerial building-footprint capture.
stats.wdc_2023_census_housing_data_by_sa2_clipped	Used to assign developable properties to suburb/SA2 reporting areas.

6. Configuration and Rule Tables

6.1 LDCAP Rules Table

1 `property.property_grz_ldcap_rules`

This table stores configurable rule parameters used throughout the LDCAP model. Values are retrieved using the helper function `property.get_rule_record(text)`.

Key Rules

Rule Name	Value	Purpose
min_building_area	60	Minimum building footprint area considered in analysis.
min_mic_radius	7	Minimum maximum-inscribed-circle radius for a viable building platform.
min_lot_size_standard	400	Standard minimum lot size.
min_frontage_general	3.6	General minimum frontage/access width.
min_frontage_large_scale	10	Minimum frontage for large-scale development.
max_building_coverage	0.25	Maximum building coverage threshold used in viability screening.
min_frontage_quadrant	15	Minimum frontage length for front-quadrant viability.
min_lot_open_space	30	Minimum open-space requirement.
pct_multi_zone_dominance	40	Minimum GRZ dominance threshold for multi-zoned properties.
min_side_access	3.6	Minimum side access width for infill viability.
min_building_distance	3	Buffer distance from existing buildings.
pct_large_scale_efficiency	70	Yield efficiency factor for large-scale developments.

6.2 Overlay Analysis Configuration Table

1 property.property_grz_ldcap_overlay_analysis_config

This table defines which spatial overlays are included in the dynamic overlay analysis. Each row identifies a source schema, table, alias, optional filter, and whether the overlay contributes to the combined constraint geometry.

Examples include:

1 Designations
2 Open Space Zone
3 Local Facilities
4 Building Consents
5 High Tension Powerline Setbacks
6 Flood Risk Areas
7 Land Stability Areas
8 Depression Areas
9 Slope >= 20
10 High Liquefaction

The overlay analysis procedure dynamically creates output columns for each configured overlay using the configured alias values.

6.3 Exclusion Thresholds Table

1 property.property_grz_ldcap_exclusion_thresholds
--

This table defines percentage or numeric thresholds used to reject properties from viable vacant or infill outputs. It supports operators including:

1 >
2 >=
3 <
4 <=
5 =
6 BETWEEN

The table also stores the source overlay-analysis column used for the threshold comparison.

7. Utility Functions

7.1 `property.get_rule_record(text)`

Returns configured rule values from `property.property_grz_ldcap_rules`.

```
1 SELECT *  
2 FROM property.get_rule_record('min_lot_size_standard');
```

Returns:

```
1 value_numeric  
2 value_text  
3 value_bool
```

This function centralises rule retrieval and allows rule values to be changed without rewriting the main SQL logic.

7.2 `general.calc_overlap_stats_record(...)`

Calculates overlap between an input geometry and a dynamically specified target table. It returns:

```
1 overlap_geom  
2 overlap_area  
3 overlap_pct
```

The function validates geometries, calculates intersections, dissolves overlaps, and reports overlap area as a percentage of the input geometry.

Used primarily by:

```
1  
property.create_mv_property_grz_ldcap_viability_overlay_analysis()
```

to dynamically process multiple constraint layers.

7.3 `property.property_buildings_ranking(integer, integer)`

Ranks buildings on a property to identify likely primary dwellings or important structures. Ranking considers:

- Building area
- Distance to frontage
- Alignment with frontage
- Back-of-property penalty

- Likely dwelling classification

The function returns ranked building geometry, building position, frontage-distance geometry, and scoring metrics.

8. Materialized Views

8.1 Building Quadrant Statistics

View

```
1 property.property_grz_ldcap_building_quadrant_stats
```

Created By

```
1 property.create_mv_property_grz_ldcap_building_quadrant_stats()
```

Purpose

Splits each GRZ property into four quadrants and analyses building presence, building overlap, and road frontage per quadrant.

Method

1. Select GRZ properties.
2. Calculate property centroid and bounding box.
3. Create vertical and horizontal split lines through the centroid.
4. Split the property into quadrants.
5. Classify each quadrant as NE, NW, SE, or SW.
6. Calculate road frontage length per quadrant.
7. Clip building footprints to property geometry.
8. Calculate building overlap per quadrant.
9. Determine whether a quadrant is buildable.

Key Outputs

Field	Description
prop_no	Property identifier.
quadrant	Quadrant label: NE, NW, SE, SW.
quadrant_id	Numeric quadrant identifier.
frontage_length	Length of road frontage associated with the quadrant.
quadrant_is_front_facing	Indicates whether the quadrant has the greatest frontage.

is_buildable_quadrant	True where the quadrant has frontage and no qualifying building.
quadrant_building_count	Number of qualifying buildings in the quadrant.
quadrant_total_building_area	Total building area associated with the quadrant.

8.2 Building Quadrant Summary

View

```
1 property.property_grz_ldcap_building_quadrant_summary
```

Created By

```
1
property.create_mv_property_grz_ldcap_building_quadrant_summary()
```

Purpose

Aggregates quadrant statistics to the property level.

Key Outputs

```
1 total_buildings
2 total_building_area
3 ne_count
4 nw_count
5 se_count
6 sw_count
7 buildable_quadrant_count
8 front_facing_quadrant_id
9 max_frontage_length
10 total_frontage_length
11 property_geom
```

This summary is later used by infill viability testing, particularly for identifying front-facing buildable quadrants.

8.3 Viability Overlay Analysis

View

```
1 property.property_grz_ldcap_viability_overlay_analysis
```

Created By

```
1  
property.create_mv_property_grz_ldcap_viability_overlay_analysis()
```

Purpose

Calculates overlay constraint exposure, developable area, vacancy status, and road frontage for each GRZ property.

Processing Logic

The procedure dynamically builds overlay area and percentage columns from `property.property_grz_ldcap_overlay_analysis_config`. It then calculates a combined `overlay_geom`, subtracts it from the property geometry, and produces `developable_geom`, `developable_area`, and `developable_pct`.

Key Outputs

```
1 prop_area  
2 prop_overlap_area_<alias>  
3 prop_overlap_pct_<alias>  
4 prop_overlap_area_building_footprints_2025_geq60  
5 prop_overlap_pct_building_footprints_2025_geq60  
6 overlay_geom  
7 overlay_area_total  
8 overlay_pct_total  
9 developable_geom  
10 developable_area  
11 developable_pct  
12 total_road_frontage  
13 is_vacant  
14 geom
```

Vacancy is determined using building footprint intersection logic and the minimum building area rule.

8.4 GRZ LDCAP Overlays

View

```
1 property.property_grz_ldcap_overlays
```

Created By

```
1 property.create_mv_property_grz_ldcap_overlays()
```

Purpose

Identifies properties affected by development-density overlays. Currently, the script includes the north_west_structure_plan_overlay with an overlay minimum lot size of 800.

Key Outputs

```
1 prop_no
2 layer
3 min_lot_size_overlay
4 geom
```

This view is used when calculating potential lots for vacant and infill properties.

8.5 Vacant Viability Testing

View

```
1 property.property_grz_ldcap_vacant_viability_testing
```

Created By

```
1
property.create_mv_property_grz_ldcap_vacant_viability_testing()
```

Purpose

Identifies vacant GRZ properties that pass initial spatial and policy viability testing.

Exclusion Criteria

Properties are excluded if they meet any of the following conditions:

- Intersect local facilities.
- Intersect Open Space Zone.
- Are not GRZ-dominant in multi-zone analysis.
- Are below the applicable minimum lot size.
- Have multi-property centroid conflicts.
- Have significant property overlap.
- Have recent dwelling building consents.
- Are listed in the manual vacant-property exclusion rule.

Buildability Test

The process subtracts overlay-restricted areas from the property and applies:

```
1 ST_MaximumInscribedCircle()
```

The property passes if:

```
1 MIC radius >= min_mic_radius
```

and sufficient open space remains.

8.6 Vacant Viable

View

```
1 property.property_grz_ldcap_vacant_viable
```

Created By

```
1 property.create_mv_property_grz_ldcap_vacant_viable()
```

Purpose

Converts tested vacant properties into final vacant development-capacity records.

Potential Lot Calculation

Potential lots are calculated from developable area:

```
1 potential_lots = floor(developable area / applicable minimum  
lot size)
```

The applicable lot size is either:

- Overlay minimum lot size; or
- Standard minimum lot size.

PHU Yield Rules

Scenario	PHU Yield
total_potential_lots = 1	1
total_potential_lots BETWEEN 2 AND 6	Full potential lot count
total_potential_lots >= 7 and frontage meets large-scale threshold	floor(total_potential_lots * 70%)
total_potential_lots >= 7 but frontage below large-scale threshold	6
Otherwise	0

The view only includes properties with sufficient frontage and at least one potential lot.

8.7 Infill Viability Testing

View

```
1 property.property_grz_ldcap_infill_viability_testing
```

Created By

```
1  
property.create_mv_property_grz_ldcap_infill_viability_testing()
```

Purpose

Identifies already-developed GRZ properties that may support additional residential infill development.

Exclusion Criteria

Infill properties are excluded if they:

- Intersect local facilities.
- Intersect Open Space Zone.
- Are not GRZ-dominant.
- Are below twice the applicable minimum lot size.
- Have building coverage greater than or equal to 25%.
- Have multi-property centroid conflicts.
- Have significant property overlap.
- Have multiple buildings and high building coverage.
- Have recent dwelling consents.
- Have no existing buildings.
- Are manually excluded using the infill exclusion rule.

Spatial Processing

The infill pipeline:

1. Clips building footprints to the property.
2. Removes small buildings below the minimum building area.
3. Measures building setback distances to side boundaries.
4. Determines whether compliant side access exists.
5. Removes constrained overlay areas.
6. Removes existing buildings plus a minimum building-distance buffer.
7. Applies a maximum-inscribed-circle buildability test.
8. Applies additional access and front-quadrant checks.

Acceptance Logic

A property can pass if it has:

- At least one compliant side-access path; or
- Very low building footprint overlap; or
- A buildable front-facing quadrant with sufficient frontage.

It must also satisfy the MIC radius and open-space requirements.

8.8 Infill Viable

View

```
1 property.property_grz_ldcap_infill_viable
```

Created By

```
1 property.create_mv_property_grz_ldcap_infill_viable()
```

Purpose

Converts infill testing results into final infill development-capacity records.

Yield Logic

Infill yield assumes an existing dwelling remains, so the PHU yield represents additional dwelling capacity only.

Scenario	PHU Yield
total_potential_lots = 1	0
total_potential_lots BETWEEN 2 AND 6	total_potential_lots - 1
total_potential_lots >= 7 and frontage meets large-scale threshold	$\text{floor}((\text{total_potential_lots} - 1) * 70\%)$
total_potential_lots >= 7 but frontage below large-scale threshold	6
Otherwise	0

The view requires sufficient frontage and more than one potential lot.

8.9 Fully Developed Testing

View

```
1 property.property_grz_ldcap_fully_developed_testing
```

Created By

```
1 property.create_mv_property_grz_ldcap_fully_developed_testing()
```

Purpose

Identifies GRZ properties that are not excluded and are not classified as vacant or infill during testing. The view also calculates a maximum-inscribed-circle platform geometry for review.

Outputs

```
1 prop_no
2 potential_land_development_type
3 radius
4 platform_area
5 platform_geom
6 geom
```

The development type is recorded as Developed.

8.10 Fully Developed

View

```
1 property.property_grz_ldcap_fully_developed
```

Created By

```
1 property.create_mv_property_grz_ldcap_fully_developed()
```

Purpose

Classifies remaining GRZ properties as fully developed where they are not vacant viable, not infill viable, and not removed by key exclusions.

Output Logic

Fully developed records are assigned:

```
1 potential_land_development_type = 'Fully Developed'
2 phu_yield = 0
```

The view also includes land area, total area, total road frontage, and geometry.

8.11 District Summary

View

```
1 property.property_grz_ldcap_summary
```

Created By

```
1 property.create_mv_property_grz_ldcap_summary()
```

Purpose

Provides district-wide summary statistics for vacant and infill development capacity.

Development Categories

Vacant records are categorised as:

```
1 Single Vacant Lot
2 Small Scale Development
3 Large Scale Development
```

Infill records are categorised as:

```
1 Single Lot Infill
2 Small Scale Infill
3 Large Scale Infill
```

The summary reports property counts, developable lots, PHU yield, and total land area.

8.12 Suburb Summary

View

```
1 property.property_grz_ldcap_summary_suburbs
```

Created By

```
1 property.create_mv_property_grz_ldcap_summary_suburbs()
```

Purpose

Provides suburb-level summaries by assigning each developable property to a suburb/SA2 area using ST_PointOnSurface(d.geom) and the SA2 clipped census geometry.

Outputs

```
1 suburb_name
```

```

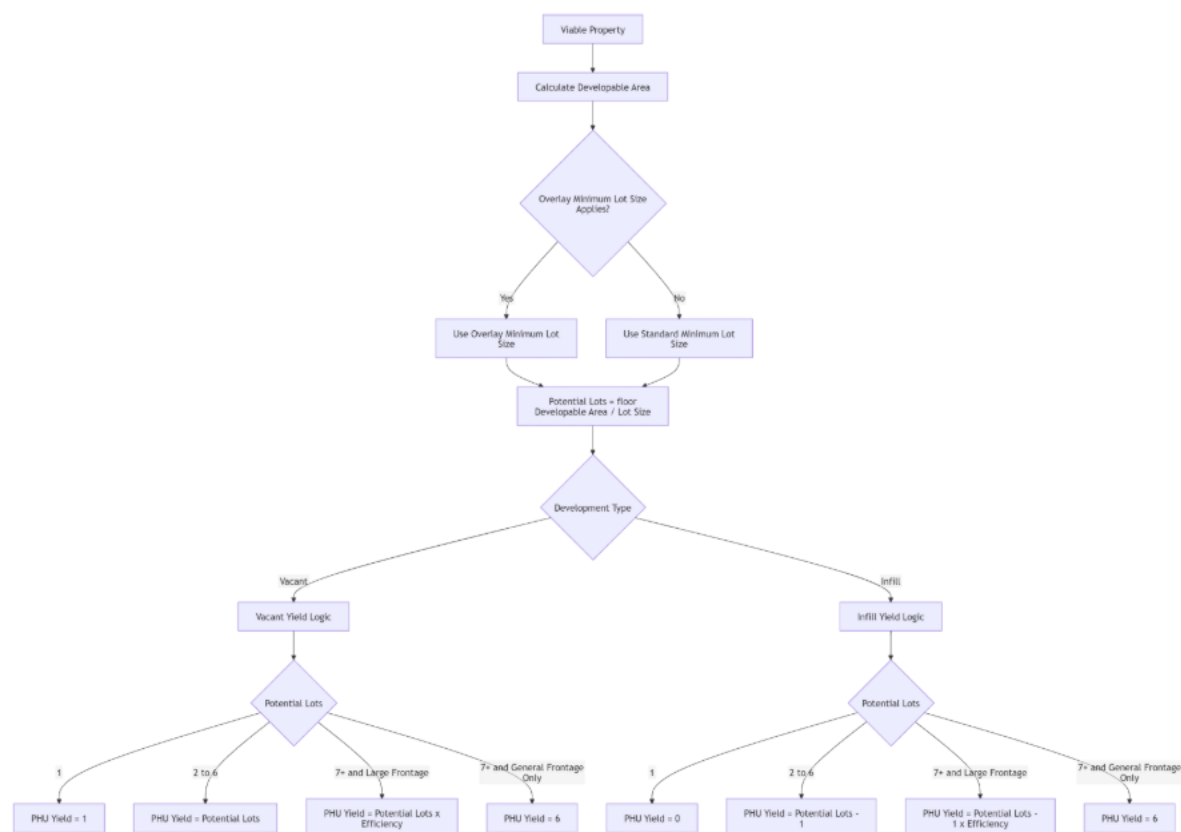
2 potential_land_development_type
3 development_category
4 developable_properties
5 total_developable_lots
6 total_phu_yield
7 total_land_area_ha

```

9. Yield Methodology

9.1 Yield Calculation Logic

The diagram below illustrates the different yield rules for vacant and infill properties. Vacant land counts the full modelled residential capacity, while infill land deducts one lot/dwelling to account for the assumed existing dwelling. Large-scale capacity is adjusted by the configured efficiency factor.



9.2 Vacant Yield

Vacant properties count the full expected residential capacity because no existing dwelling is assumed to occupy the site.

1 Vacant PHU Yield = potential lots

For large-scale development, the model applies an efficiency factor to account for land required for internal access, roading, layout inefficiencies, or subdivision design.

9.3 Infill Yield

Infill properties assume one existing dwelling remains.

1 Infill PHU Yield = potential lots - 1

Large-scale infill development also applies the configured efficiency factor.

10. Spatial Methods Used

The script relies heavily on PostGIS functions, including:

1 ST_Intersects
2 ST_Intersection
3 ST_Difference
4 ST_Union
5 ST_UnaryUnion
6 ST_Collect
7 ST_Area
8 ST_Length
9 ST_Buffer
10 ST_MaximumInscribedCircle
11 ST_PointOnSurface
12 ST_Centroid
13 ST_Envelope
14 ST_Split
15 ST_Dump
16 ST_DumpSegments
17 ST_Boundary
18 ST_ShortestLine
19 ST_ClosestPoint
20 ST_LineLocatePoint

```
21 ST_Within
22 ST_Transform
23 ST_MakeValid
```

These are used to evaluate zoning, hazards, buildable area, property overlap, frontage, side access, building coverage, and development platform feasibility.

11. Indexing Strategy

Each materialized view generally receives:

- A fid btree index.
- A prop_no btree index.
- A geometry GiST index.
- Additional geometry GiST indexes where multiple geometry columns exist.

Examples include indexes on:

```
1 geom
2 building_geom
3 quadrant_geom
4 platform_geom
5 overlay_geom
6 developable_geom
7 property_geom
```

This indexing supports spatial querying, map display, filtering, and downstream analysis.

12. Permissions and Ownership

Objects are generally owned by:

```
1 postgres
```

Permissions are granted to:

```
1 postgres
2 wdc
3 wdc_gisprod
4 PUBLIC
```

Materialized views generally grant full access to postgres and wdc, while wdc_gisprod receives SELECT access. Stored procedures and functions generally grant EXECUTE to PUBLIC, postgres, wdc, and wdc_gisprod.

13. Regeneration and Audit Comments

Each materialized view is annotated with a regenerated timestamp using:

1 COMMENT ON MATERIALIZED VIEW ...

The comment records that the view was regenerated by the relevant stored procedure and includes LOCALTIMESTAMP.

This provides a lightweight audit trail for identifying when a materialized view was last rebuilt.

14. Known Technical Considerations

14.1 Full Rebuild Strategy

The master procedure drops and recreates all materialized views rather than refreshing them in place.

Implication:

This ensures consistent dependency ordering but may require downtime or careful scheduling if views are used by production services.

14.2 Dynamic SQL Dependency

The overlay-analysis view is generated using dynamic SQL based on rows in property_grz_ldcap_overlay_analysis_config.

Implication:

Adding or changing configuration rows can alter the output schema of property_grz_ldcap_viability_overlay_analysis.

14.3 Manual Exclusion Lists

Vacant and infill processes both use text-based rule values containing comma-separated property numbers.

Implication:

Manual exclusions should be documented and reviewed regularly to avoid hidden model assumptions.

14.4 Overlay Threshold Column Mapping

The exclusion-threshold process depends on source_column values matching dynamically generated overlay-analysis column names.

Implication:

If overlay aliases change, corresponding threshold mappings must also be updated.

15. Recommended Operational Procedure

Rebuild All Outputs

```
1 CALL property.create_mv_property_grz_ldcap();
```

Validate Outputs

Recommended post-run checks:

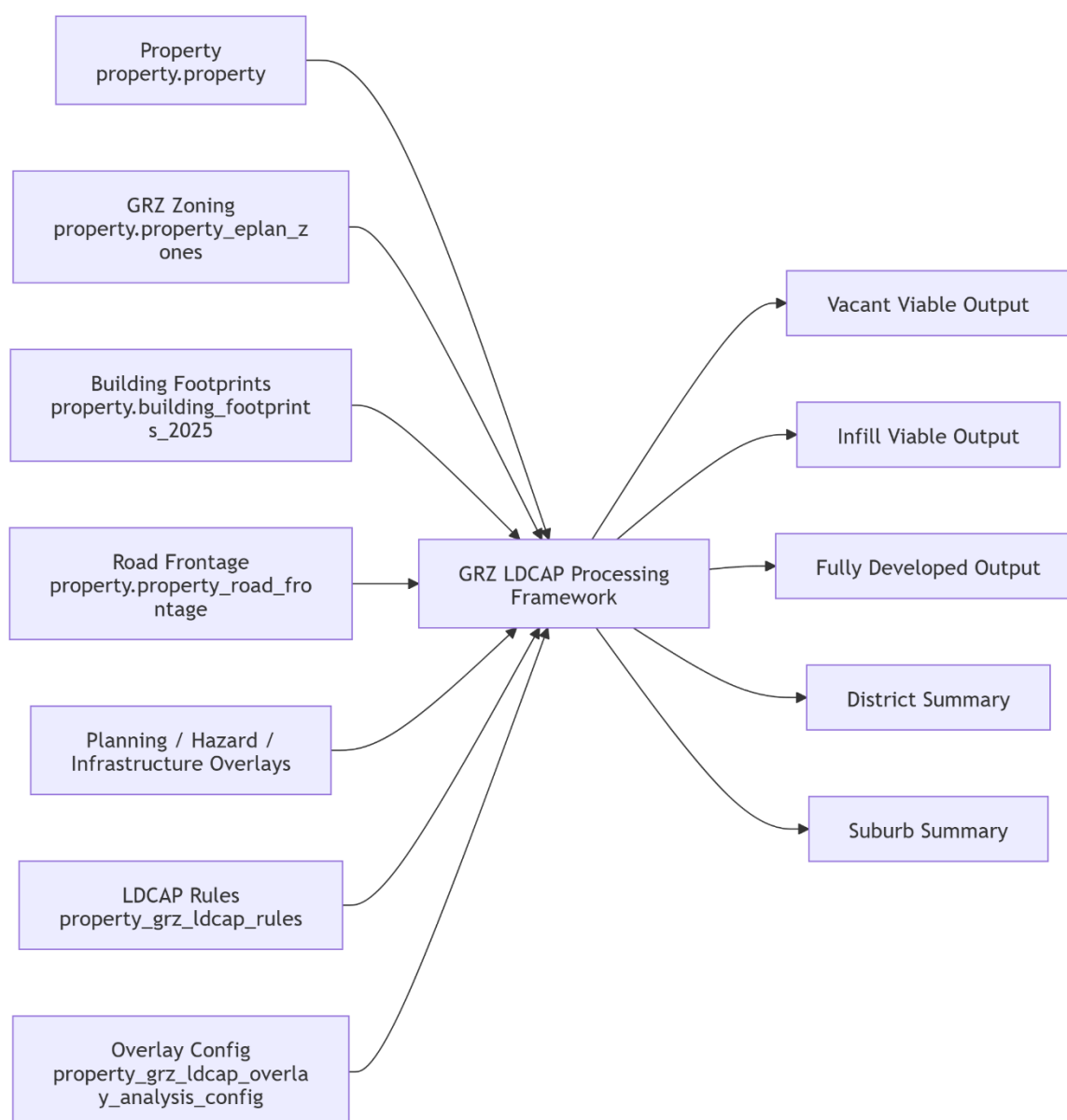
```
1 SELECT COUNT(*)
2 FROM property.property_grz_ldcap_vacant_viable;
3
4 SELECT COUNT(*)
5 FROM property.property_grz_ldcap_infill_viable;
6
7 SELECT COUNT(*)
8 FROM property.property_grz_ldcap_fully_developed;
9
10 SELECT *
11 FROM property.property_grz_ldcap_summary
12 ORDER BY potential_land_development_type,
development_category;
```

Review Last Regeneration Timestamp

```
1 SELECT obj_description(
2     'property.property_grz_ldcap_summary'::regclass,
3     'pg_class'
4 );
```

16. Data Flow Diagram

This diagram shows the main source datasets flowing into the LDCAP framework and the principal output products generated by the SQL process. The framework combines property, zoning, building, frontage, overlays, configuration, and rules data to produce vacant, infill, fully developed, and summary outputs.



17. Output Catalogue

Output	Type	Description
property_grz_ldcap_building_quadrant_stats	Materialized view	Detailed quadrant/building/frontage analysis.
property_grz_ldcap_building_quadrant_summary	Materialized view	Property-level quadrant summary.
property_grz_ldcap_viability_overlay_analysis	Materialized view	Overlay, developable area, frontage, and vacancy analysis.
property_grz_ldcap_overlays	Materialized view	Density overlay property mapping.
property_grz_ldcap_vacant_viability_testing	Materialized view	Vacant property buildability testing.
property_grz_ldcap_vacant_viable	Materialized view	Final vacant development capacity output.
property_grz_ldcap_infill_viability_testing	Materialized view	Infill property buildability testing.
property_grz_ldcap_infill_viable	Materialized view	Final infill capacity output.
property_grz_ldcap_fully_developed_testing	Materialized view	Developed property testing output.
property_grz_ldcap_fully_developed	Materialized view	Final fully developed property output.
property_grz_ldcap_summary	Materialized view	District-wide capacity summary.
property_grz_ldcap_summary_suburbs	Materialized view	Suburb-level capacity summary.

18. Glossary

Term	Definition
GRZ	General Residential Zone.
LDCAP	Land Development Capacity.
PHU Yield	Potential Household Unit yield.
MIC	Maximum Inscribed Circle, used to test whether a viable building platform can fit within remaining developable land.
Infill	Additional development on an already-developed property.
Vacant	Development opportunity on a property treated as having no qualifying existing building.
Fully Developed	Property not identified as vacant viable or infill viable.

Overlay	Spatial constraint or planning control intersecting a property. Overlay areas are used to determine developable land and, where required, apply alternative development standards such as minimum lot size requirements.
Developable Area	Property area remaining after relevant overlay constraints are removed.